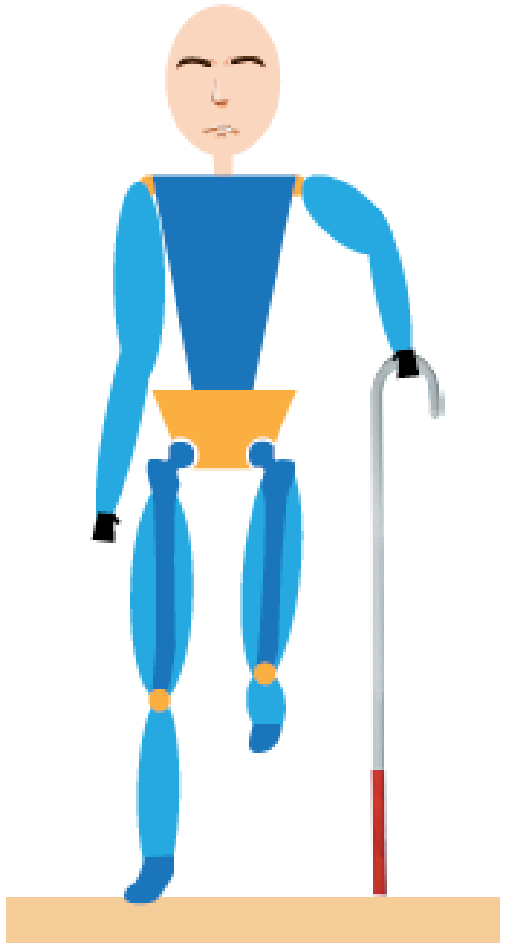
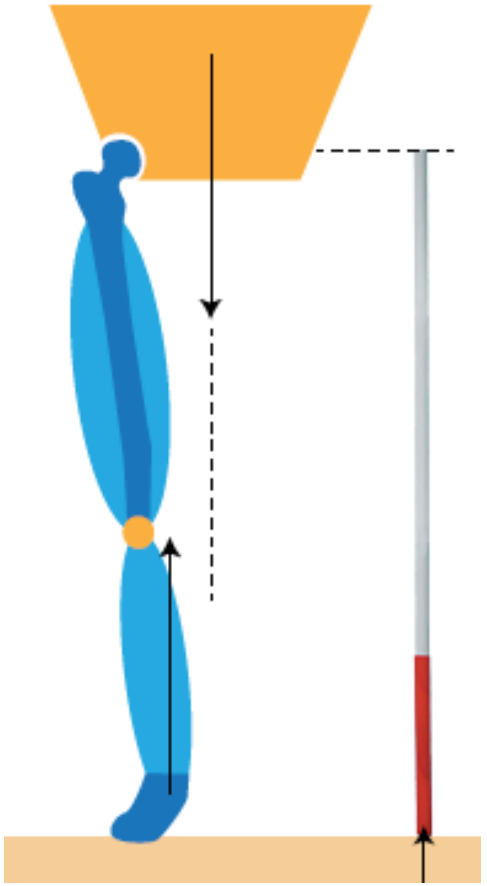
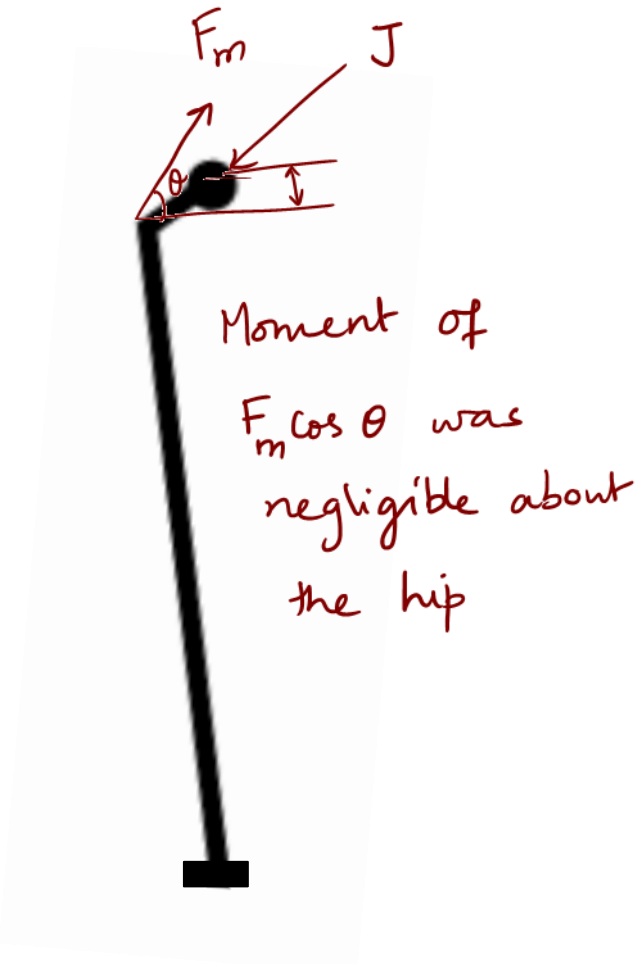
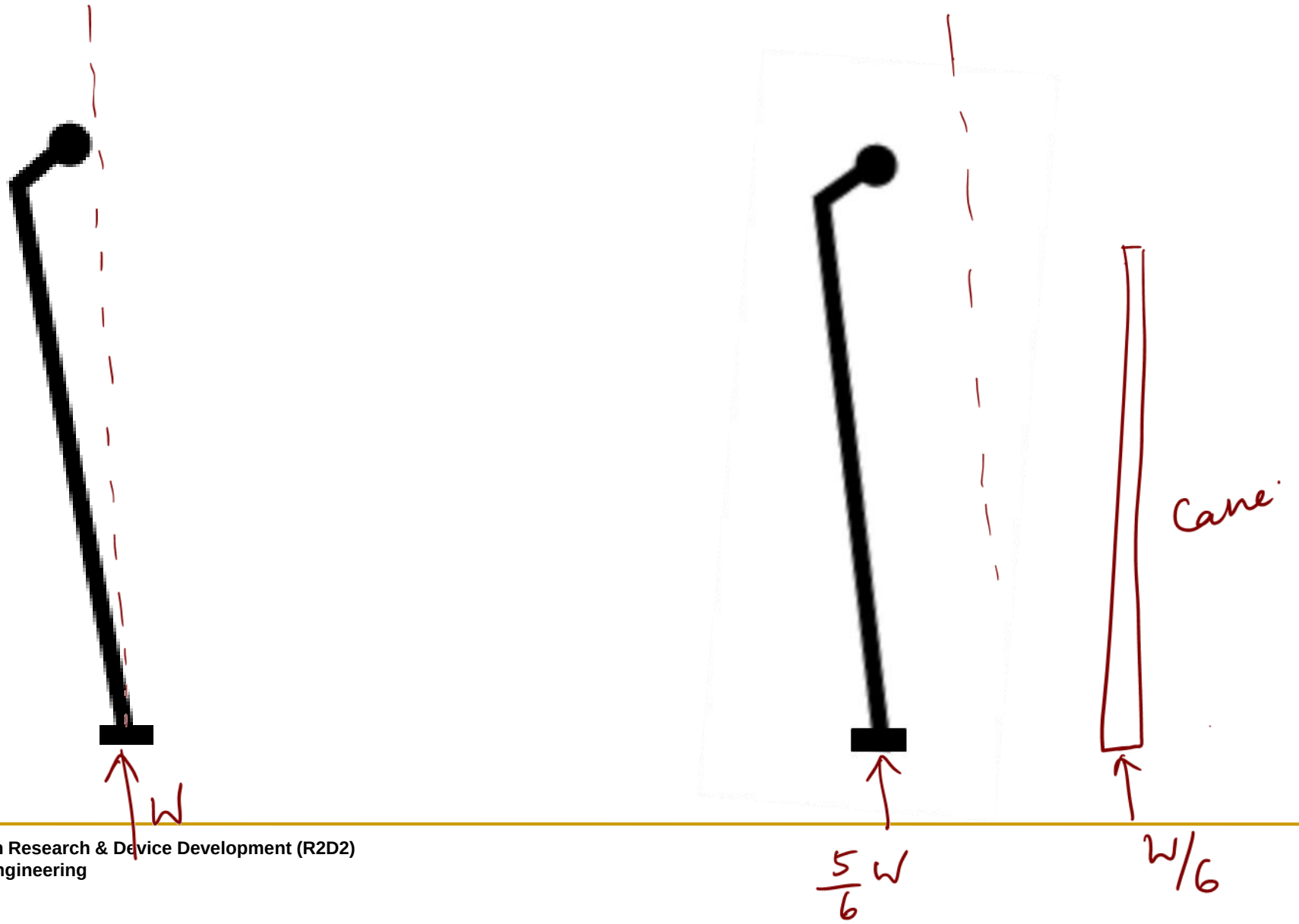
Static analysis – using a cane



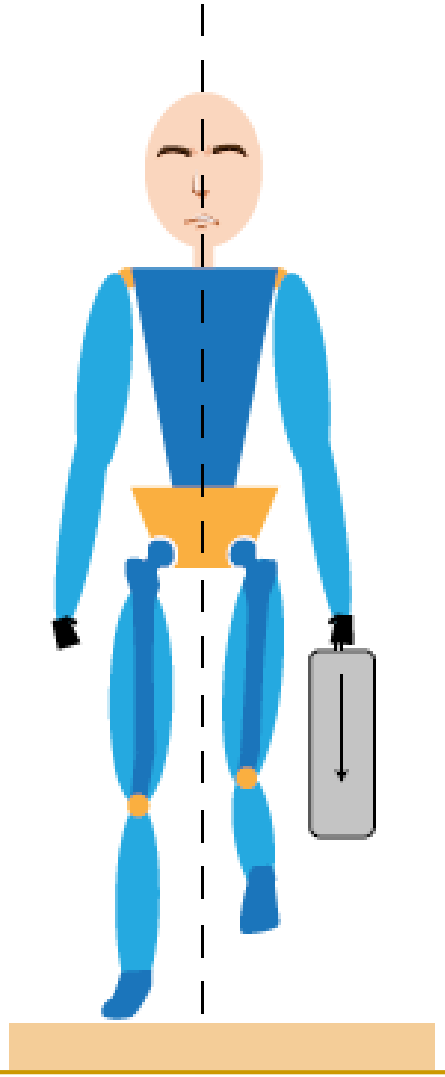


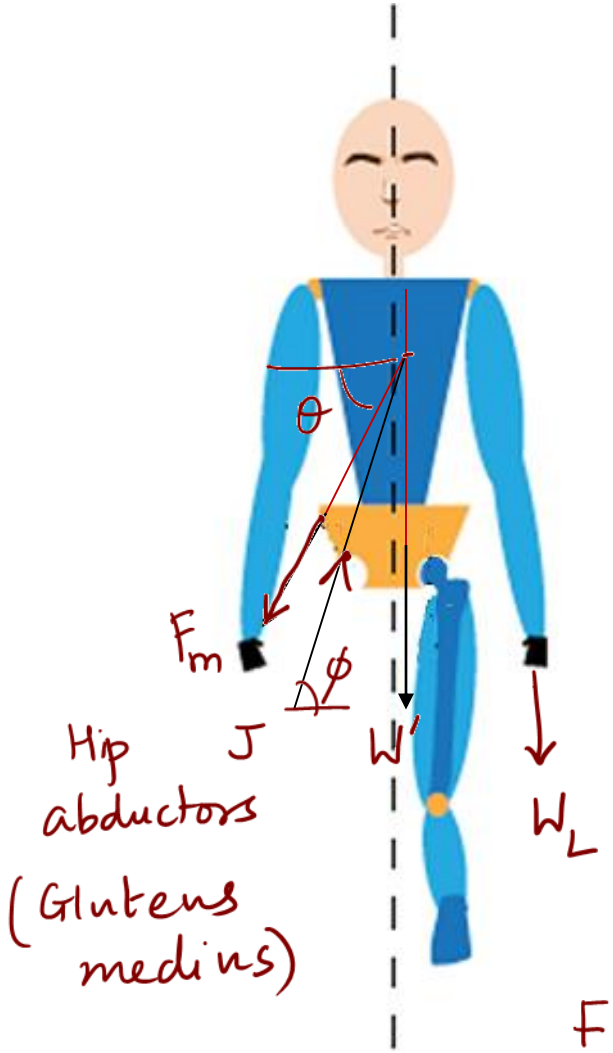


Static analysis – single stance vs using a cane



Carrying a load on one side



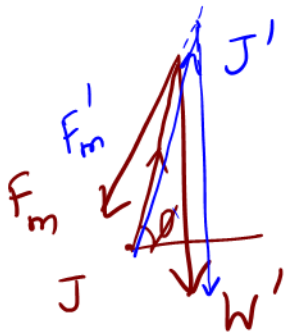


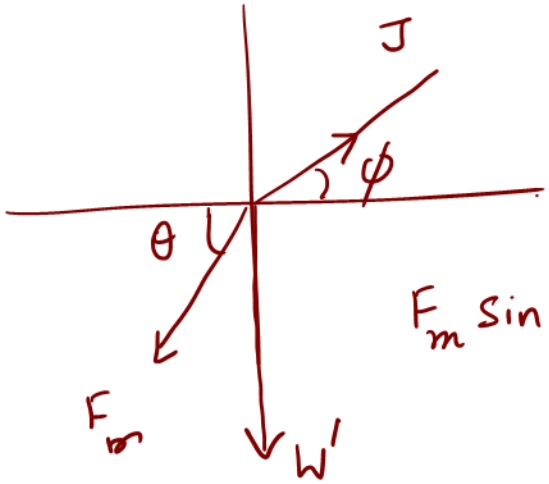
supporting leg

$W' = W - W_{leg}$

$W'' = W - W_{leg} + W_L$

If $W_L = 0$



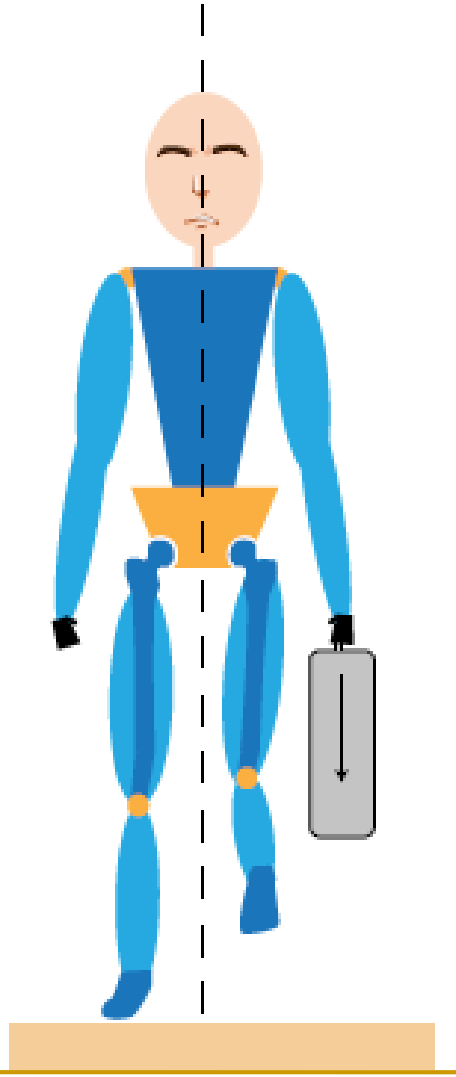


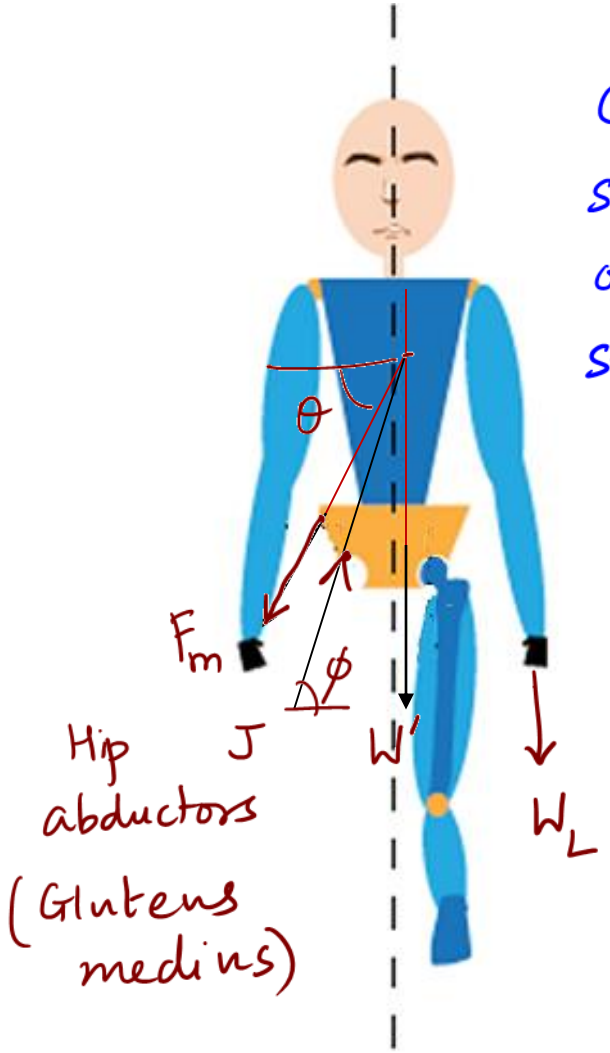
$F_m \cos\theta = J \cos\phi$

$F_m \sin\theta + W' = J \sin\phi$

$$F_m = \frac{W' \cos\phi}{\cos\theta \sin\phi - \sin\theta \cos\phi} = \frac{W' \cos\phi}{\sin(\phi - \theta)}$$

Carrying a load on one side

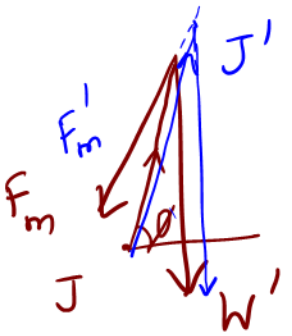


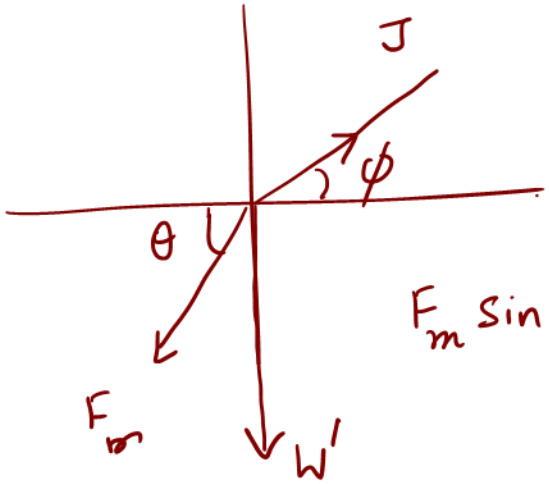


Carrying a load on one side increases the hip loads on the opposite side single stance.

supporting leg

$W'' = W - W_{leg} + W_L$



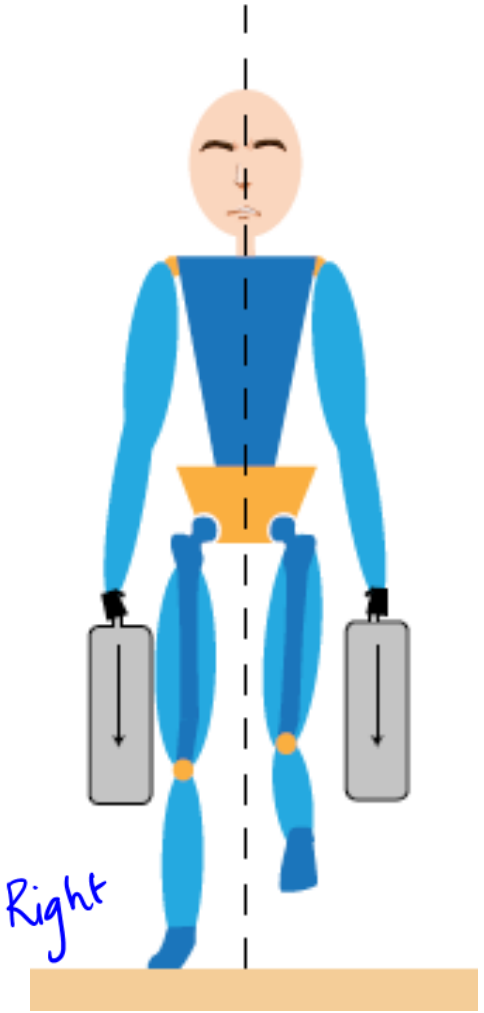


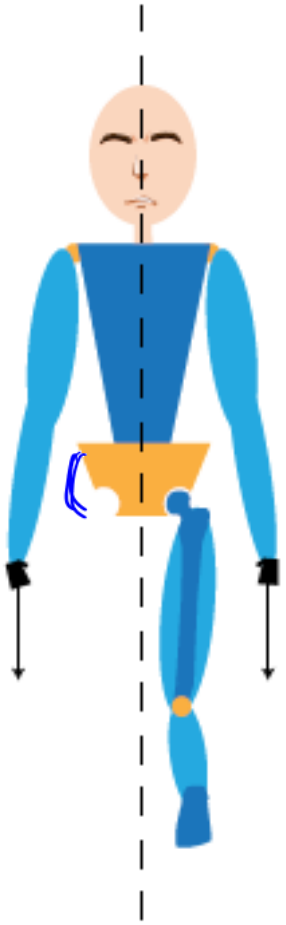
$F_m \cos\theta = J \cos\phi$

$F_m \sin\theta + W' = J \sin\phi$

$$F_m' = \frac{W'' \cos\phi'}{\cos\theta \sin\phi' - \sin\theta \cos\phi'} = \frac{W'' \cos\phi'}{\sin(\phi' - \theta)}$$

Carrying a load equally distributed on both sides

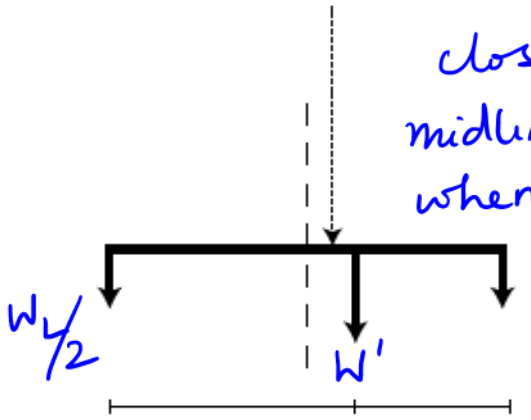




$W' = W - W_{leg}$

$W'' = W' + W_L$

closer to the midline than when carrying on one side



Bringing loads closer to the midline of the body helps to reduce the musculoskeletal loads

When carrying a load on one side, we lean towards the other side

This brings the CG of the upper body and loads being carried closer to the midline of the body and reduces the length of the moment arm from the farther hip joint

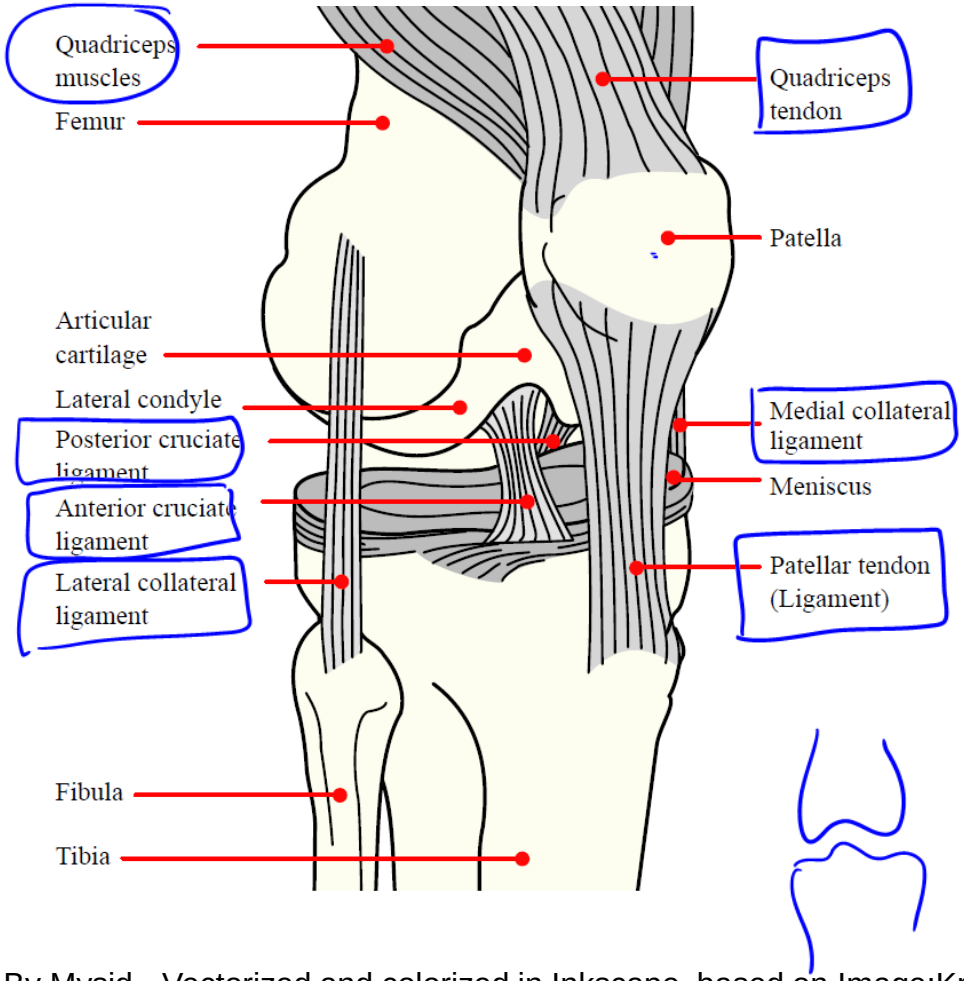
People with weak hip abductor muscles and/or painful hip joints — they lean towards the loading side (in single stance) — "abductor gait"

Weak left side — lean towards right during right single stance

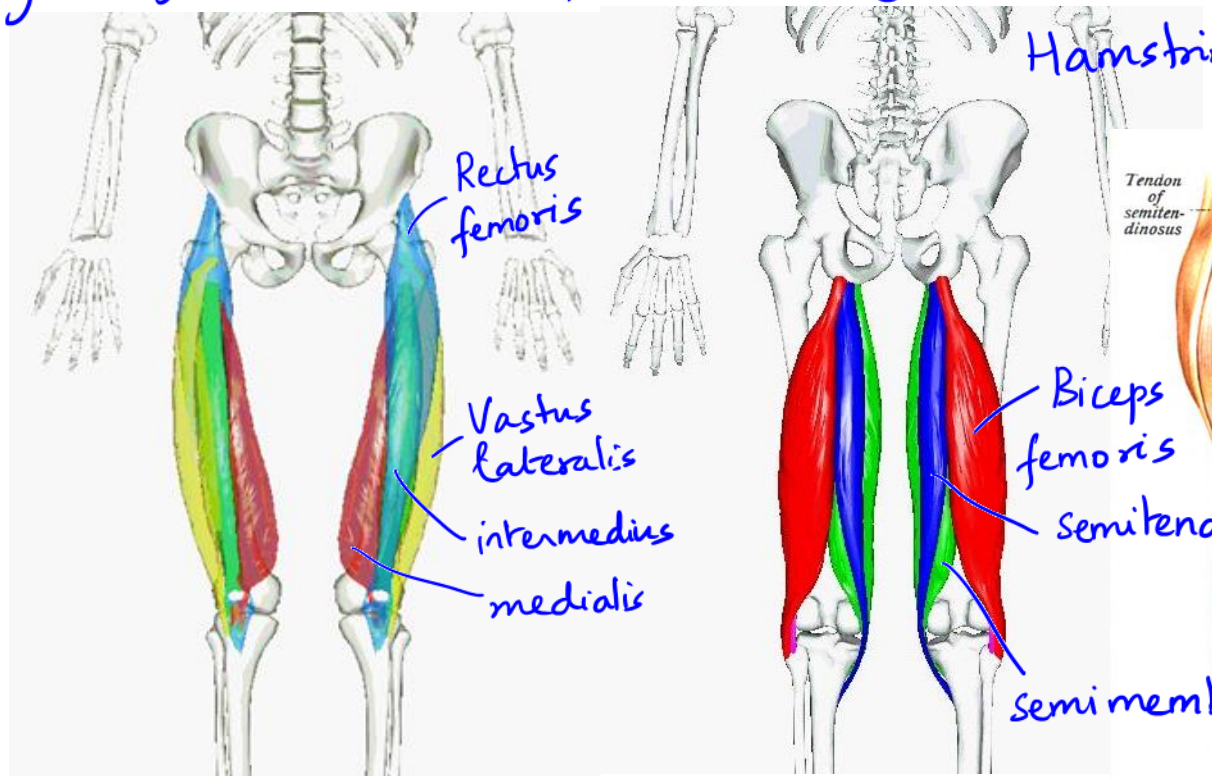
and vice-versa

Anatomy – knee

Condyloid joint – modified hinge joint



By Mysid - Vectorized and colorized in Inkscape, based on Image:Knee diagram.png., Public Domain, https://commons.wikimedia.org/w/index.php?curid=4004042



By BodyParts3D/Anatomography - BodyParts3D/Anatomography, CC BY-SA 2.1 jp, https://commons.wikimedia.org/w/index.php?curid=33808194

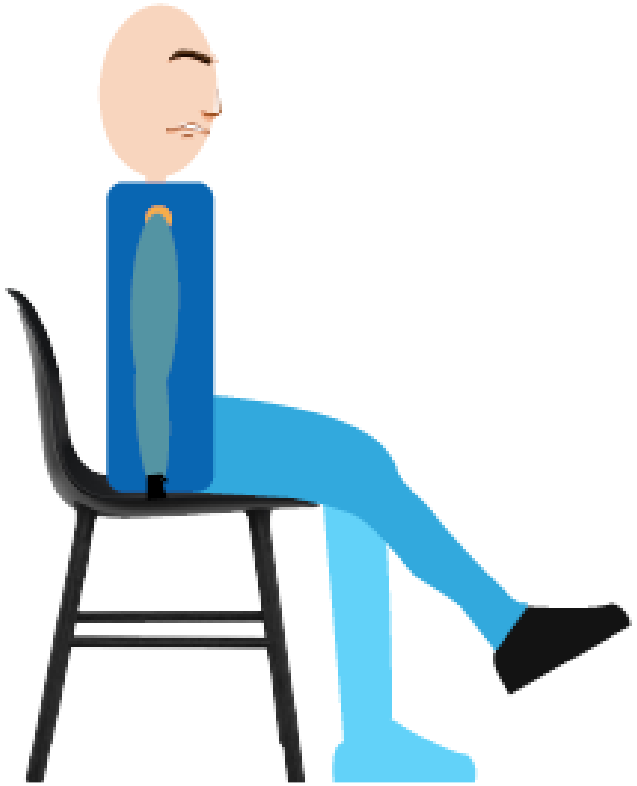
Tibiofemoral joint – hinge joint

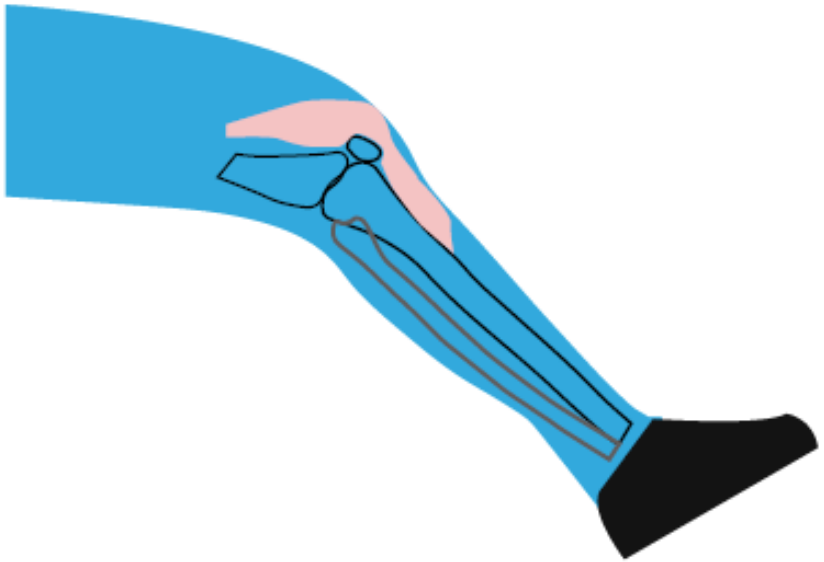
Patellofemoral joint – knee cap



By Dr. Johannes Sobotta - Sobotta's Atlas and Text-book of Human Anatomy 1909, Public Domain, https://commons.wikimedia.org/w/index.php?curid=29822860

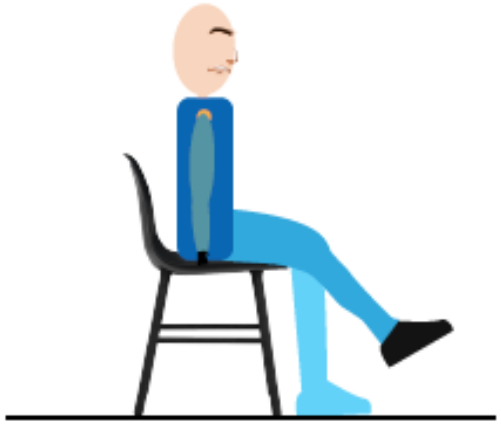
Static analysis - knee

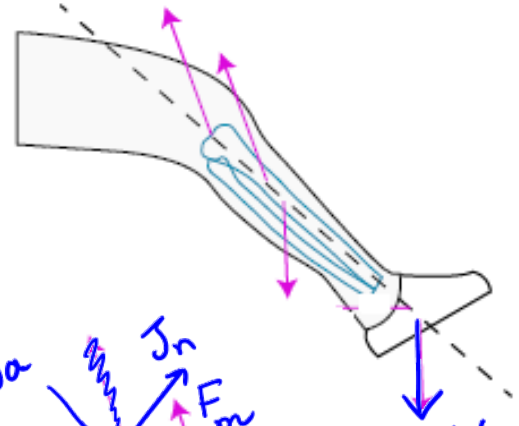




Static analysis - knee

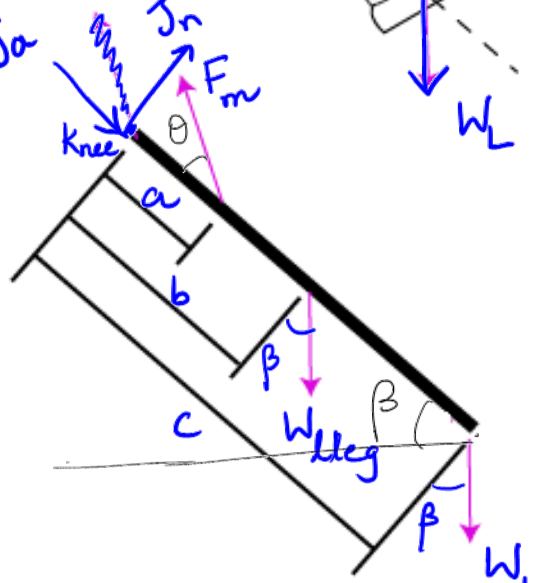
Person wearing ankle weights doing flexion/extn exercises from a seated position to strengthen the quadriceps muscle





$$\Sigma M_{knee} = 0$$

$$(F_m \sin\theta)\, a - W_{lleg} \cos\beta \cdot b - W_L \cos\beta \cdot c = 0$$



$$J_n + F_m \sin\theta - W_{lleg} \cos\beta - W_L \cos\beta = 0$$

$$J_a - F_m \cos\theta + W_{lleg} \sin\beta + W_L \sin\beta = 0$$

Note correction

$a = 12$ cm, $b = 22$ cm, $c = 50$ cm, $\theta = 15°$, $\beta = 45°$

$W_{lleg} = 150$ N

$W_L = 100$ N

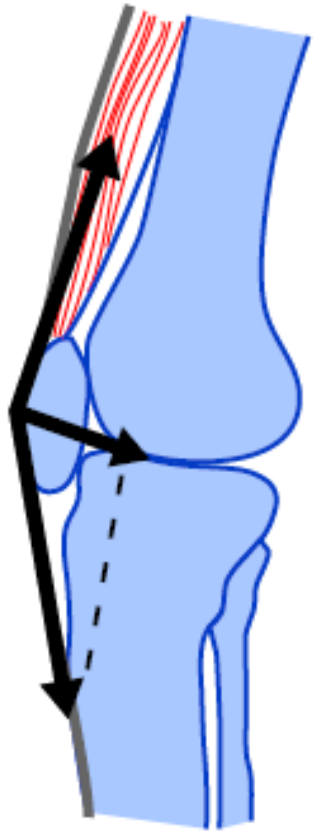
$F_m = 1890$ N

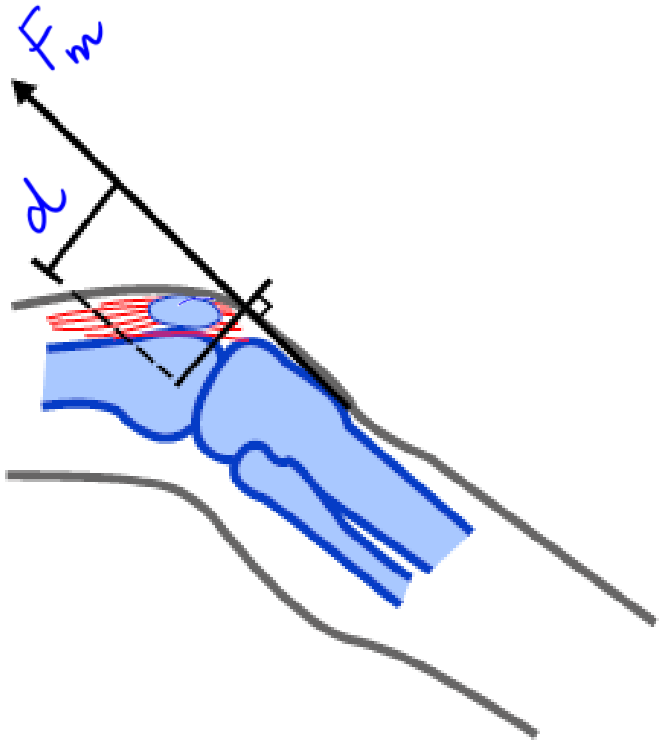
$J_n = 312.4$ N , $J_a = -1648.8$ N

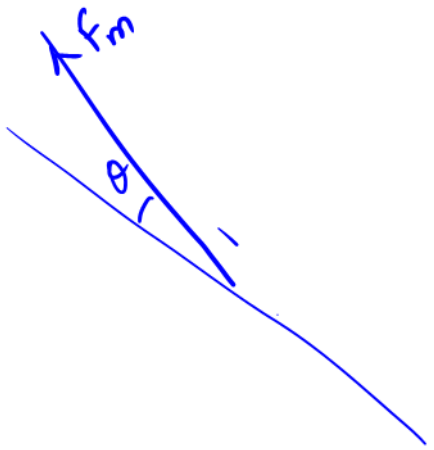
$\therefore J = \sqrt{J_a^2 + J_n^2}$

$= 1678$ N

Role of patella







If patella is removed,
d reduces

Patellar tendon increases the lever arm of the knee extensors by increasing θ